

# **Project Synopsis**

for

## **Identity Security Framework for Supply Chain**

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## **I. Title of the Project**

Identity Security Framework for Supply Chain.

## **II. Statement of the Problem**

Logistics networks suffer from a lack of "Identity Visibility," where cargo data is often stored in plain text and accessible via weak credentials. This exposes the supply chain to data tampering and cargo theft. The problem is to implement a project that not only secures the identity of stakeholders (Suppliers, Managers) but also protects the operational data (Content, Origin) using industrial-grade encryption.

## **III. Why this particular topic chosen?**

Supply chain security is a global priority. This topic allows for the exploration of "Zero-Trust" data principles in a logistics context. It is technically challenging as it requires the integration of cryptographic libraries (Fernet) with real-time mapping technologies (Leaflet.js) to show how identity drives the movement of goods in a safe environment. The problem was chosen not because it is easy, but because it bridges the gap between domain-specific logistics needs and modern cybersecurity practices. Future scope includes live GPS tracking and biometric MFA.

## **IV. Objective and Scope**

- **Objective:** To build an industrial command center where every logistical asset is encrypted and every user is multi-factor authenticated (Google OAuth + TOTP). The system enforces role-based access control (RBAC) for Suppliers, Logistics Coordinators, and Managers.
- **Scope:** Includes Supplier dispatching, Logistics management, and Managerial oversight. The scope covers the full lifecycle of cargo from "In Transit" to "Delivered," secured by OAuth and TOTP. The project does not cover payment processing or real vehicle tracking

## V. Methodology

A **Hybrid Waterfall-Agile Methodology** was used. The security requirements (encryption, authentication, RBAC) were defined upfront following a Waterfall approach, as these are non-negotiable. The interactive dashboards and mapping telemetry were developed in iterative 2-week Agile sprints, allowing rapid UI feedback. This hybrid model ensures both security rigor and user-centered design.

## VI. Process Description

The system consists of three main modules:

1. **Identity Hub** – Handles user registration, login (password + Google OAuth), TOTP setup/verification, and session management.
2. **Encryption Layer** – Uses Fernet (symmetric encryption) to automatically encrypt cargo fields (content, origin, destination) before database insertion via SQLAlchemy ORM.
3. **Logistics Hub** – Displays a real-time Leaflet.js map with geo-telemetry of shipments; role-based dashboards show different data views.

**Process Logic:** When a Supplier dispatches cargo, the data is encrypted by the SecurityLayer before hitting the database. Only authenticated and 2FA-verified users can decrypt and view the cargo details. The flow of information is supported by Level-0 DFD and a Login Sequence Flowchart.

## VII. Resources and Limitations

- **Hardware:** Development PC (8GB+ RAM), high-speed internet for map tile rendering and OAuth callbacks.
- **Software:** Python 3.10+, Flask, Flask-Login, SQLite/PostgreSQL, SQLAlchemy, cryptography (Fernet), authlib (Google OAuth), pyotp (TOTP), Leaflet.js, HTML/CSS.
- **Operating System:** Windows 11 / Linux (Ubuntu) for development; any OS with a modern browser for client.

- **Limitations:** The system relies on static city-to-coordinate mapping for the map display; live GPS integration is future work. The MFA is TOTP-based (no hardware tokens or biometrics). The system is web-only, no mobile application.

## **VIII. Testing Technologies used**

- **Unit Testing (White-box):** 20 test cases covering password hashing, TOTP generation, Fernet encryption/decryption, and role checks. Conducted during development using pytest.
- **Integration Testing (Black-box):** 15 test cases for OAuth redirect flow, 2FA verification after login, and database encryption hooks. Conducted after module integration in Week 6.
- **User Interface Testing:** Cross-browser verification (Chrome, Firefox) of the Leaflet map and dashboard layouts. 10 test cases for role-based menu visibility.
- **System Integration Test:** Matching Supplier dispatch IDs with Logistics update actions to ensure audit trail integrity. Conducted before final submission.
- **Schedule:** Unit tests – Week 5; Integration tests – Week 6; UAT – Week 7; Security validation – Week 8.

## **IX. Conclusion**

Kinetic Forge demonstrates that logistical efficiency does not have to come at the cost of security. By intertwining Field-Level Encryption (Fernet) with Multi-Factor Authentication (TOTP + Google OAuth), the system provides a robust blueprint for modern, anti-hack logistical platforms. The innovation lies in the seamless fusion of real-time mapping with enterprise-grade identity security, making it stand out from traditional supply chain dashboards that often ignore data-at-rest protection.



